Multi-Hop Question Answering using Retrieval-Augmented Generation (RAG)

Objective

The objective of this project was to build a baseline Retrieval-Augmented Generation (RAG) system for multi-hop question answering using the HotpotQA dataset. Instead of focusing solely on benchmark performance, the project emphasizes understanding where the model succeeds, where it fails, and how retrieval quality affects final answer generation.

Dataset Selection

I selected the **HotpotQA Question Answering Dataset**, specifically the **hotpot_dev_distractor_v1.json** split.

This dataset is well suited for evaluating Retrieval-Augmented Generation because:

- Questions require reasoning across multiple Wikipedia articles.
- Ground-truth answers are available for evaluation.
- Supporting facts are provided, allowing retrieval analysis.
- The distractor setting provides a realistic retrieval challenge.

Baseline Method

The baseline RAG pipeline consists of the following components:

1. Load and preprocess HotpotQA documents.
2. Build a document corpus from the provided contexts.
3. Generate dense document embeddings using **Sentence Transformers (all-MiniLM-L6-v2)**.
4. Index document embeddings using **FAISS**.
5. Retrieve the Top-5 most relevant documents for each question.
6. Generate answers using **FLAN-T5 Base** conditioned on the retrieved context.
7. Evaluate predictions using Exact Match (EM), F1 Score, and Retrieval Accuracy.

Results

Metric	Score
Exact Match	0.36
F1 Score	0.48
Retrieval Accuracy	0.99

The retriever successfully identified relevant supporting documents for most questions. However, the overall answer quality remained lower than the retrieval accuracy, indicating that answer generation and multi-hop reasoning remain the primary challenges.

Failure Analysis

Several common failure patterns were observed:

- Partial answers where only part of the required information was generated.
- Hallucinated responses despite retrieving relevant documents.
- Multi-hop reasoning failures requiring information from multiple documents.
- Occasional retrieval misses for complex questions.

The large gap between Retrieval Accuracy (0.99) and Exact Match (0.36) suggests that the retrieval component performs well, while the generation model struggles to combine evidence across multiple retrieved documents.

Improvement

As a potential improvement, I explored increasing the retrieval depth from Top-5 to Top-10 retrieved documents. The goal was to provide additional context to the language model for complex multi-hop questions. Due to time constraints, the focus of this submission is the baseline system and its evaluation. Future work would compare the baseline and improved retrieval settings using the same evaluation metrics.

Future Work

Possible future improvements include:

- Using stronger embedding models such as **BAAI/bge-base-en-v1.5**.
- Applying a cross-encoder reranker after retrieval.
- Improving prompt engineering.
- Using larger instruction-tuned language models.
- Fine-tuning the retriever on HotpotQA.